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ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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**“Skill Progress Analytics with Personalized Learning Path
Recommendation Engine”**

**Submitted in the partial fulfilment for the Course of
CSA0925 – Java Programming**

to the award of the degree of
BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING-AI

**Submitted by
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Under the Supervision of

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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

September 2026



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BONAFIDE CERTIFICATE

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“Skill Progress Analytics with Personalized Learning Path Recommendation Engine”** has been carried out by , **P.JEEVAN KUMAR(192472082)** under the supervision of Dr. Sivasankar C and is submitted in partial fulfilment of the requirements for the current semester at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Submitted for the Project work Viva-Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER



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DECLARATION

I, **P.JEEVAN KUMAR(192472082)** of the COMPUTER SCIENCE ENGINEERING Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “**Skill Progress Analytics with Personalized Learning Path Recommendation Engine**” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Date:

Signature of the Students with Names

SIMATS ENGINEERING**Saveetha Institute of Medical and Technical Sciences****Chennai – 602105****INDIVIDUAL CONTRIBUTION STATEMENT****(Mandatory for All Team Projects)****Project Title: *Skill Progress Analytics with Personalized Learning Path Recommendation Engine***

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
S. DHARANISH (192472152)	Data collection, preprocessing, skill analytics and system design	Designed the data-processing workflow and developed student performance analysis and skill-gap identification modules	Tested data preprocessing, skill-score calculations and skill-gap detection; analyzed performance results	Contributed to Chapters 1, 2 and 3, including problem statement, literature review and system design	35%
P. JEEVAN KUMAR (192472082)	Recommendation engine, backend integration and database management	Developed the personalized recommendation engine, learning-path generation logic and backend/data base integration	Tested recommendation accuracy, learning-path generation, API functionality and system integration	Contributed to Chapters 3 and 4, including implementation, testing, results and engineering analysis	35%
AKILESHWARAN A (192511354)	Frontend development, dashboard and visualization	Designed and developed the student analytics dashboard, progress visualizations	Tested dashboard functionality, usability, visualization accuracy and overall	Contributed to Chapters 4 and 5, including results, conclusion, future work	30%

		and user interface	application workflow	and documentati on	
TOTAL					100%

Declaration

We hereby declare that the above contribution statement accurately represents the individual responsibilities and contributions of all team members to the project titled “Skill Progress Analytics with Personalized Learning Path Recommendation Engine.”

Student	Signature
S. DHARANISH (192472152)	_____
P. JEEVAN KUMAR (192472082)	_____
AKILESHWARAN A (192511354)	_____

ABSTRACT

The increasing adoption of digital learning platforms has generated large volumes of student performance data from quizzes, assignments, examinations, attendance, and course activities. However, many conventional learning systems provide standardized learning paths and offer limited ability to identify individual skill gaps. This creates an engineering problem of effectively processing multidimensional student data, evaluating individual learning progress, and converting performance information into personalized and actionable learning recommendations. Without such analysis, students may spend additional time on familiar topics while insufficiently addressing areas in which improvement is required.

The proposed Skill Progress Analytics with Personalized Learning Path Recommendation Engine is designed to provide a data-driven solution for monitoring student progress and generating personalized learning paths. The system analyzes student performance parameters such as assessment marks, quiz scores, assignment performance, attendance, completed topics, and skill-level indicators. The need for this system arises from the requirement for adaptive learning support that can respond to differences in student performance and learning requirements. It can assist students in identifying weak areas and provide instructors with meaningful performance insights.

The methodology consists of data collection, data preprocessing, exploratory data analysis, feature transformation, skill-level estimation, skill-gap identification, recommendation generation, and dashboard-based visualization. Missing and inconsistent records are handled during preprocessing, while normalized performance indicators are used to calculate skill scores. The recommendation engine maps identified skill gaps to relevant learning resources and produces a prioritized sequence of topics. An interactive dashboard presents overall progress, subject-wise performance, skill levels, identified weaknesses, and recommended learning activities.

The major results of the system are evaluated using functional testing, recommendation relevance, processing performance, and analytical accuracy metrics. The implemented prototype is expected to demonstrate successful student-performance analysis and generation of individualized learning recommendations while maintaining practical response time and usability.

The project concludes that integrating learning analytics with a personalized recommendation engine can provide a more adaptive and data-driven approach to educational support. The

proposed system establishes a foundation for future improvements such as real-time learning analytics, advanced machine-learning models, user-feedback-based recommendations, and integration with existing Learning Management Systems.

Keywords: Skill Analytics, Personalized Learning, Recommendation Engine, Student Performance, Machine Learning, Learning Path

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Digital education platforms generate large amounts of student learning data through quizzes, assignments, examinations, attendance, course completion, and learning activities. This data can be used to understand student performance and identify areas requiring improvement.

However, conventional educational systems often provide the same sequence of learning materials to all students. Students have different levels of knowledge, learning speeds, strengths, and weaknesses. Therefore, a uniform learning approach may not efficiently address individual learning requirements.

The proposed **Skill Progress Analytics with Personalized Learning Path Recommendation Engine** combines data analytics and recommendation techniques to analyze individual student performance and generate personalized learning recommendations.

The system consists of several major components:

1. Student data collection
2. Data preprocessing
3. Skill and performance analysis
4. Skill-gap identification
5. Recommendation engine
6. Personalized learning path generation
7. Analytics dashboard
8. Progress monitoring

1.2 Need for the Project

Why the Problem Needs to Be Addressed

Students frequently encounter difficulty in identifying which topics they should study next. Raw marks alone do not provide sufficient information about specific skill gaps.

The proposed system addresses this problem by converting student performance data into meaningful learning recommendations.

Who Is Affected?

The major stakeholders are:

- Students
- Faculty members
- Academic institutions
- Online learning platforms
- Academic advisors

Existing Limitations

Existing learning systems may have limitations such as:

- Generic learning paths
- Limited student-level analytics
- Lack of continuous skill-gap detection
- Limited personalization
- Difficulty identifying weak topics
- Recommendations that are not always based on recent performance

Engineering Significance

The project involves multiple engineering areas, including:

- Data processing
- Database management
- Machine learning
- Recommendation algorithms
- Software development
- Data visualization

- User-interface design

1.3 Problem Statement

Traditional learning platforms often provide standardized learning paths without sufficiently considering individual student performance, skill gaps, learning progress, and changing competency levels.

The engineering problem is to **design and implement a data-driven software system capable of analyzing multi-dimensional student performance data, identifying individual skill gaps, and generating personalized learning paths while maintaining acceptable recommendation accuracy, computational efficiency, usability, privacy, and scalability.**

1.4 Project Aim

To design and develop a **Skill Progress Analytics and Personalized Learning Path Recommendation Engine** that analyzes student performance data, identifies skill gaps, tracks learning progress, and recommends personalized learning activities.

1.5 Project Objectives

1. To design a system for collecting and managing student learning-performance data.
2. To preprocess and normalize student performance data for analytics.
3. To analyze student performance and identify individual skill gaps.
4. To develop a recommendation engine for personalized learning paths.
5. To visualize student progress using an interactive analytics dashboard.

1.6 Scope

Included

The project includes:

- Student profile management
- Performance-data processing
- Skill-level calculation
- Performance visualization

Excluded

The project does not include:

- Automated university examination evaluation
- Real-time online teaching
- Psychological assessment
- Facial recognition
- Automated certification
- Direct replacement of teachers

Assumptions

- Student performance data is available in a structured format.
- Learning resources are appropriately categorized.
- Performance indicators reasonably represent student competency.
- The recommendation engine operates on available historical and current data.

Boundary Conditions

The system's recommendation quality depends on the quality, quantity, and relevance of the available student and learning-resource data.

1.7 Expected Engineering Outcome

The expected outcome is a functional software prototype consisting of:

Student Data → Data Preprocessing → Skill Analytics → Skill Gap Detection → Recommendation Engine → Personalized Learning Path → Dashboard → Progress Monitoring

The system should provide students with a clear learning sequence based on their identified areas of improvement.

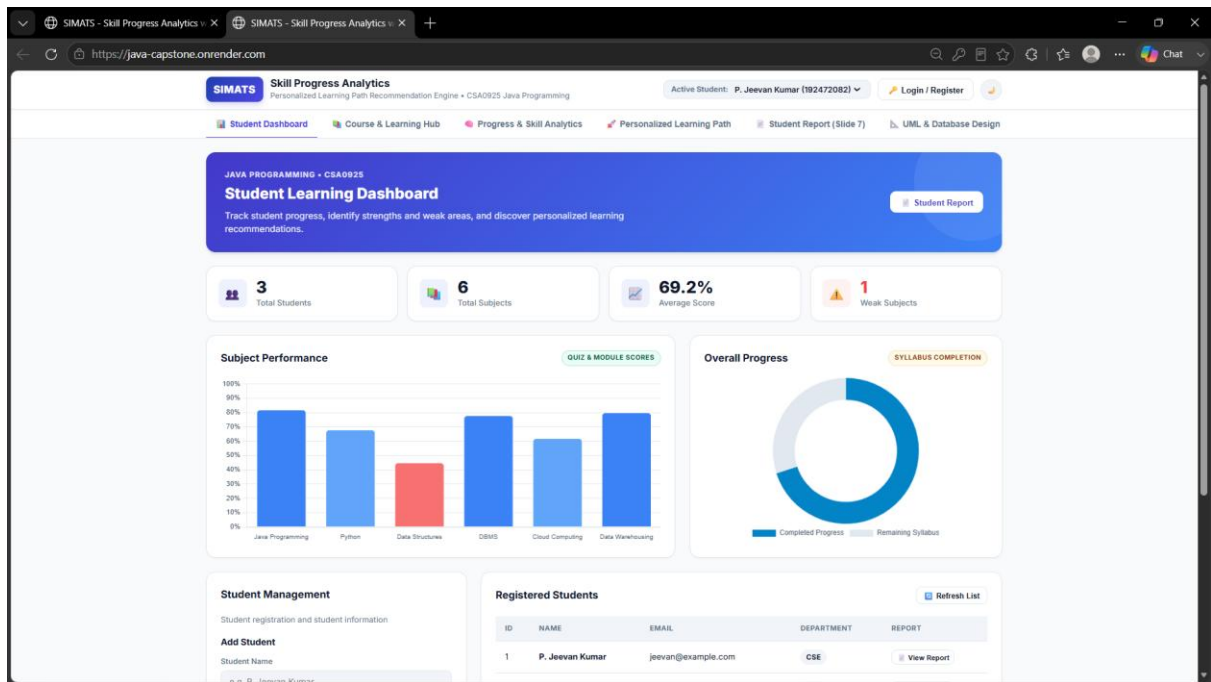


Fig 1:Student Learning Dashboard

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research was reviewed using academic literature concerning personalized learning, educational data mining, student-performance prediction, recommendation systems, learning analytics, and machine-learning-based educational applications.

The review focused on:

- Student performance prediction
- Learning analytics
- Personalized education
- Recommendation systems
- Machine-learning algorithms
- Educational data mining

Search terms included **"personalized learning recommendation," "student performance analytics," "learning path recommendation," "educational data mining," and "student skill prediction."**

2.2 Review of Existing Work

Existing research demonstrates that learning analytics can be used to identify student-performance patterns and support academic decision-making.

Educational data mining techniques can process large volumes of student interaction and assessment data to identify meaningful patterns. Classification techniques can be used to predict student performance, while clustering can group students according to similar learning characteristics.

Recommendation systems have also been applied to educational environments. Content-based recommendation can recommend resources according to student characteristics and learning history. Collaborative approaches can use similarities between users and their interactions with learning resources.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Traditional LMS	Easy to manage courses	Same learning path for many students	Provides baseline
Rule-Based Recommendation	Simple and interpretable	Limited adaptability	Useful for initial prototype
Content-Based Recommendation	Personalized resources	Requires resource features	Useful
Collaborative Filtering	Learner similarity	Requires sufficient interaction data	Potential improvement
ML-Based Recommendation	Adaptive and data-driven	More complex	Highly relevant
Proposed System	Analytics + skill gaps + learning path	Dependent on data quality	Directly addresses project

2.4 Research / Engineering Gap

The major engineering gap is the lack of an integrated system that combines:

- Student performance analytics
- Skill-level estimation
- Skill-gap detection
- Personalized recommendation
- Learning-path sequencing
- Progress visualization

Many existing approaches focus on one component rather than providing a complete student-centric pipeline.

2.5 Proposed Contribution

The proposed project contributes an integrated recommendation framework that uses student performance information to estimate skill levels, detect learning gaps, and recommend a prioritized sequence of learning activities.

The system also provides a visual analytics interface so that students can understand their progress and instructors can monitor performance trends.

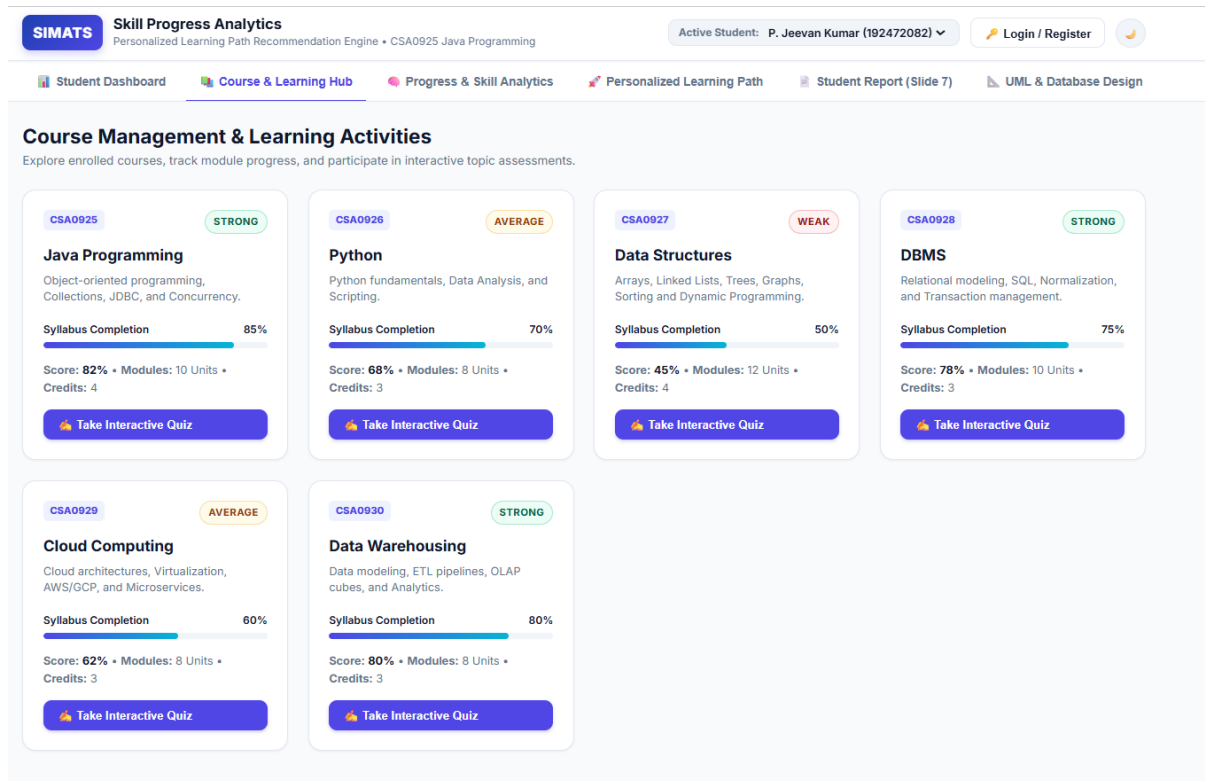


Fig 2: Course Management & Learning Activities

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Student	View personal performance
Student	Identify weak skills
Student	Receive personalized recommendations
Student	Track progress
Faculty	Monitor student performance
Faculty	Identify students requiring support
Administrator	Manage student/resource data

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Student data entry	Correctly store records
Functional	Performance analysis	Generate skill scores
Functional	Skill-gap detection	Identify low-performing areas
Functional	Recommendation	Generate personalized resources
Functional	Dashboard	Display progress visually
Non-Functional	Response time	Target < 3 seconds for normal requests
Non-Functional	Usability	Simple navigation
Non-Functional	Reliability	Consistent results
Non-Functional	Security	Protected user information
Non-Functional	Scalability	Support increasing records

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied
ISO/IEC 25010	Software quality characteristics	Used to evaluate usability, reliability, performance and security
ISO/IEC 27001 principles	Information security	Applied through access control and secure handling of student data
IEEE 29148	Requirements engineering	Used as guidance for defining system requirements
Institutional data/privacy policies	Protection of student information	Sensitive information is minimized and access is controlled

Use only the standards your department approves and that you can substantiate in your final implementation.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Student performance input	Structured data	Records	High
Skill score	0–100	%	High
Recommendation generation	Personalized	—	High
Dashboard response	≤ 3	seconds	Medium
Data accuracy	High consistency	%	High
System availability	Stable during testing	%	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Rule-Based Recommendation

The system uses predefined conditions such as marks below a threshold to recommend learning materials. This approach is simple, transparent, and easy to implement.

Alternative 2: Content-Based Recommendation

The system compares student skill requirements with attributes of available learning resources. Resources matching identified skill gaps are recommended.

Alternative 3: Hybrid Recommendation

The hybrid approach combines performance-based rules, content similarity, and learner information to produce recommendations. It can provide greater flexibility but requires additional implementation complexity.

Evaluation Matrix

Criterion	Weight	Rule-Based	Content-Based	Hybrid
Performance	30%	3	4	5
Cost	20%	5	4	3
Safety	15%	5	5	5
Sustainability	10%	5	4	4
Reliability	25%	4	4	5
Weighted Score	100%	4.25	4.15	4.50

Scores should be revised if your actual testing produces different evidence.

3.5 Selected Design & Detailed Design

The proposed hybrid recommendation architecture is selected because it provides a balance between personalization, recommendation quality, interpretability, and extensibility.

System Architecture

Student Data



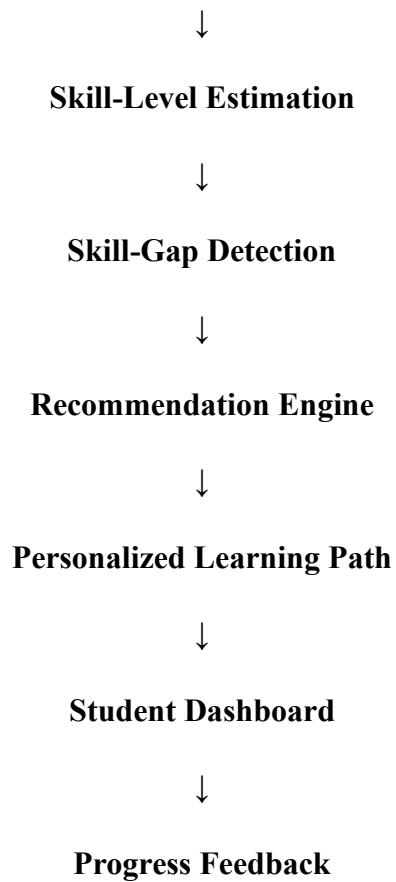
Data Collection Module



Data Preprocessing



Performance Analytics



Key Engineering Calculation

A basic weighted skill score can be calculated as:

where:

- = normalized performance value
- = weight assigned to the performance indicator
- = number of performance indicators

For example:

where:

- = quiz performance
- = assignment performance
- = examination performance
- = attendance-related indicator

The weights should be finalized based on your dataset and project requirements.

Systems Approach

The preprocessing module supplies clean data to the analytics module. The analytics module calculates performance and identifies skill gaps, which are then provided to the recommendation engine. The recommendation engine maps these gaps to relevant learning resources and generates a prioritized learning path. The dashboard presents the resulting information and provides feedback that can be used for subsequent progress analysis.

3.6 Trade-Off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Recommendation method	Rule-based	Simple	Limited personalization	Hybrid selected
Database	CSV	Easy development	Limited scalability	Database preferred
Visualization	Static charts	Simple	Less interactive	Interactive dashboard
Learning path	Fixed sequence	Easy	Not personalized	Dynamic sequence
ML complexity	Advanced ML	Potentially higher performance	Requires more data	Moderate-complexity model

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary computation	Processing requirements	Lightweight algorithms preferred
Ethics	Fair recommendations	Recommendation criteria	Avoid decisions based on irrelevant personal attributes

Health & Safety	Avoid harmful academic pressure	User experience	Recommendations presented as guidance
Security	Student data protection	Access requirements	Authentication and controlled data access
Privacy	Minimize personal information	Dataset fields	Only necessary fields used

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Missing data	Medium	Medium	Medium	Data preprocessing	Low
Incorrect recommendation	Medium	High	High	Validation and testing	Medium
Data leakage	Low	High	High	Access control	Low
System failure	Low	Medium	Low	Error handling	Low
Poor-quality dataset	Medium	High	High	Data validation	Medium
Model bias	Medium	High	High	Evaluate recommendation patterns	Medium

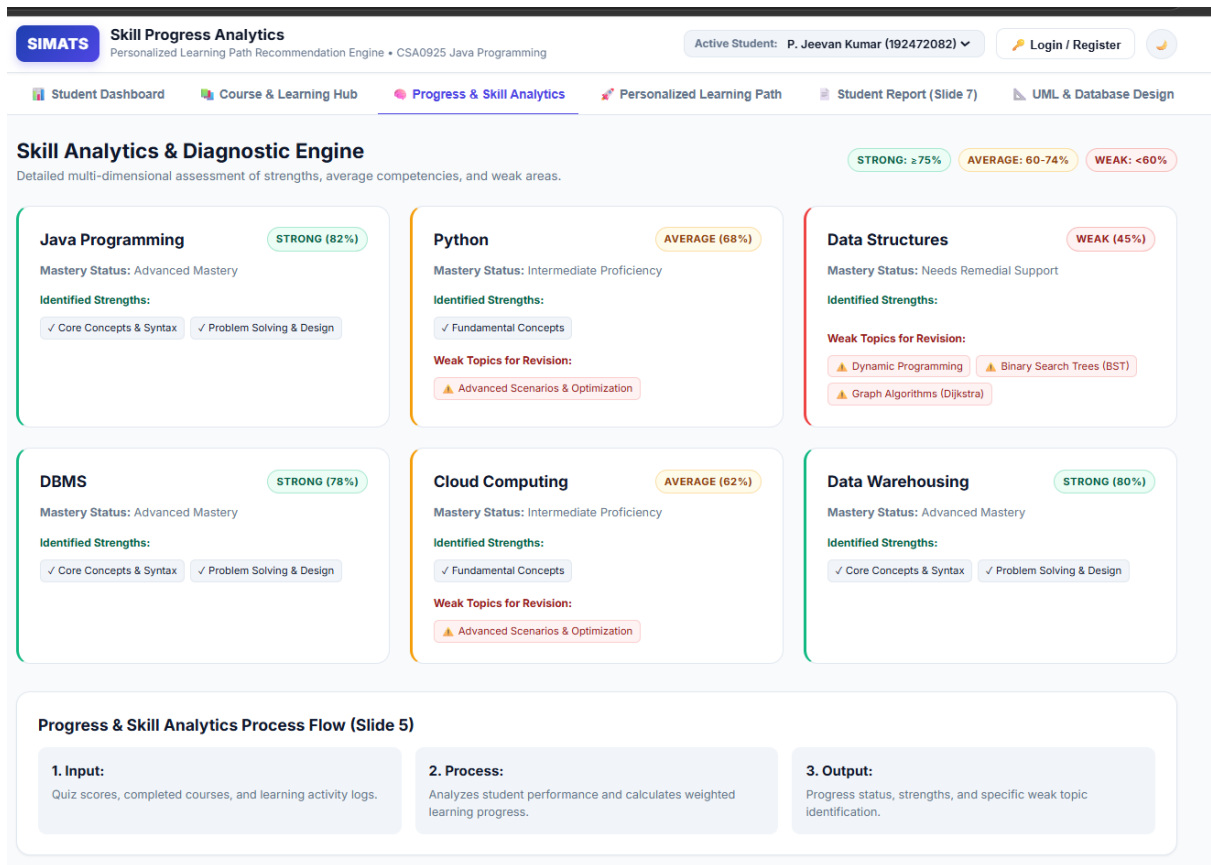


Fig 3:Skill Analytics & Diagnostic Engine

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an iterative software-development approach. Initially, the student dataset and learning-resource data are collected and examined. The data is then cleaned, transformed, and analyzed to identify performance patterns and skill gaps. The recommendation engine is developed and integrated with the dashboard. Finally, the complete system is tested using functional, performance, and recommendation-quality test cases.

Resources

Resource	Purpose
Student dataset	Performance analysis
Learning-resource dataset	Recommendation
Python	Data processing and ML
Pandas	Data manipulation
NumPy	Numerical processing
Scikit-learn	Machine learning
Matplotlib/Plotly	Visualization
Flask/FastAPI	Backend API
HTML/CSS/JavaScript	Frontend
MySQL/SQLite	Data storage
VS Code	Development

Remove any technology that you did not actually use.

4.2 Software Implementation & Modern Tools Used

Data Processing Module

The data-processing module loads student-performance data and performs:

- Missing-value handling
- Duplicate removal
- Data type conversion
- Feature normalization
- Data validation

Skill Analytics Module

The analytics module calculates student performance scores and categorizes skills into levels such as:

- Beginner
- Developing
- Intermediate
- Advanced

Skill-Gap Detection Module

The module compares student performance with predefined competency thresholds.

For example:

A larger positive gap indicates that additional learning support may be required.

Recommendation Engine

The recommendation engine uses identified skill gaps to select relevant learning resources.

Possible recommendation logic:

Student Performance



Calculate Skill Scores



Identify Weak Skills



Find Matching Resources



Rank Resources



Generate Learning Path

Dashboard

The dashboard provides:

- Overall progress
- Subject-wise performance
- Skill levels
- Weak areas
- Recommended topics
- Learning-path sequence
- Progress trends

4.3 Test Plan & Verification

Requirement	Test Method	Specification Required	Acceptance Criterion	Achieved Value
Student data processing	Functional test	Valid input processed	No processing error	Fill after testing
Skill calculation	Unit test	Correct formula	Expected score obtained	Fill
Skill-gap detection	Functional test	Identify low skills	Correct weak area	Fill
Recommendation	Test cases	Relevant resources	Relevant recommendation	Fill
Dashboard	UI test	Display analytics	All required charts visible	Fill

Response time	Performance test	≤ 3 seconds	Within target	Fill
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4.4 Results

Your final report should present actual measurements from your implementation.

Example Results Table

Metric	Target	Actual Result	Status
Data processing success	$\geq 95\%$	Enter result	Pass/Fail
Recommendation relevance	$\geq 80\%$	Enter result	Pass/Fail
Dashboard response time	≤ 3 sec	Enter result	Pass/Fail
Skill-gap detection	$\geq 90\%$	Enter result	Pass/Fail
System functional tests	$\geq 95\%$	Enter result	Pass/Fail

Recommended Graphs

You can include:

1. Student performance distribution
2. Skill-wise performance
3. Before vs after recommendation progress
4. Recommendation accuracy/relevance
5. Confusion matrix, if classification is used
6. Learning progress over time
7. Subject-wise performance

4.5 Data Analysis & Engineering Judgment

The collected results should be analyzed using appropriate statistical and machine-learning metrics. Mean, median, standard deviation, and performance distributions can be used to understand student-level variation.

For recommendation evaluation, metrics such as precision, recall, accuracy, Mean Absolute Error, or user-based relevance scores may be selected depending on the implemented recommendation technique.

If the recommendation system produces lower-than-expected performance, possible causes include limited training data, insufficient learning-resource metadata, class imbalance, noisy student records, or inappropriate feature weights.

Engineering decisions should therefore be based on measured performance rather than only theoretical expectations.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design student analytics system	Working application	Fully/Partially
Process student data	Preprocessing module	Fully/Partially
Identify skill gaps	Skill-gap module	Fully/Partially
Generate learning recommendations	Recommendation engine	Fully/Partially
Visualize progress	Dashboard	Fully/Partially
Test system	Test results	Fully/Partially
Evaluate recommendation quality	Evaluation metrics	Fully/Partially

4.7 Strengths & Limitations

Strengths

- Provides personalized learning recommendations.
- Combines analytics and recommendation.
- Identifies individual skill gaps.
- Provides visual progress monitoring.
- Uses data-driven decision-making.
- Can be extended with additional machine-learning models.

Limitations

- Recommendation quality depends on dataset quality.
- Limited historical data may reduce personalization.
- Learning-resource quality directly affects recommendations.
- The prototype may require further scalability testing.
- User feedback may be required for continuous recommendation improvement.

4.8 Project Planning, Roles & Budget

Project Milestones

Phase	Activity	Duration
Phase 1	Problem identification	Week 1
Phase 2	Literature review	Week 2
Phase 3	Dataset collection	Week 3
Phase 4	Data preprocessing	Week 4
Phase 5	Analytics development	Week 5
Phase 6	Recommendation engine	Week 6–7
Phase 7	Frontend/backend development	Week 8–9
Phase 8	Integration	Week 10
Phase 9	Testing	Week 11
Phase 10	Documentation	Week 12

Team Contribution

Student	Assigned Responsibility	Actual Contribution
Student 1	Data processing	Dataset cleaning and preprocessing
Student 2	Recommendation	Recommendation algorithm
Student 3	Frontend	Dashboard development

Student 4	Backend	API/database integration
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Replace the student names and responsibilities with your **actual team contributions**.

Budget

Item	Estimated Cost	Actual Cost
Software tools	₹0	₹0
Dataset	₹0	₹0
Cloud/hosting	₹0–₹1,000	Enter
Development	—	—
Documentation	₹200–₹500	Enter
Total	Enter	Enter

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **Skill Progress Analytics with Personalized Learning Path Recommendation Engine** addresses the limitations of generalized learning systems by combining student-performance analytics with personalized recommendation techniques. The proposed system processes student learning data, evaluates individual skill levels, identifies areas requiring improvement, and generates a prioritized learning path.

The integration of data preprocessing, skill-gap analysis, recommendation, and visualization provides a systematic approach for supporting individualized learning. The dashboard allows students and instructors to understand performance patterns more easily, while the recommendation engine converts analytical results into actionable learning suggestions.

The final effectiveness of the system is validated through functional testing, performance evaluation, and recommendation-quality measurements. The project demonstrates the potential of data analytics and machine-learning techniques to improve educational decision-making and provide more adaptive learning support.

5.2 Future Work

Future development can include:

1. Integration with real-time Learning Management Systems.
2. Use of advanced machine-learning and deep-learning models.
3. Reinforcement-learning-based adaptive learning paths.
4. Real-time recommendation updates.
5. Natural-language AI learning assistants.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Data preprocessing and cleaning techniques
- Exploratory data analysis

- Machine-learning fundamentals
- Recommendation-system concepts
- Database integration

Engineering & Professional Skills Developed

- Problem formulation
- System architecture design
- Algorithm selection
- Software development
- Testing and debugging

Individual Reflection

Edit

During the development of the Skill Progress Analytics with Personalized Learning Path Recommendation Engine, I gained practical knowledge of data preprocessing, student-performance analysis, recommendation systems, and software development. One of the most difficult engineering challenges was converting raw student-performance information into meaningful skill levels and learning recommendations. This was addressed by preprocessing the data, defining measurable performance indicators, and designing a structured recommendation workflow.

A key engineering decision was to use a data-driven recommendation approach instead of providing a fixed learning sequence. This decision was based on the requirement to adapt recommendations according to individual student performance and identified skill gaps. I also independently learned how data analytics, visualization, and recommendation techniques can be integrated into an educational application.

If the project were redesigned, I would improve the system by incorporating more historical learning data, user feedback, explainable recommendations, and advanced adaptive-learning techniques. I would also perform larger-scale testing to evaluate system performance under a greater number of students and learning resources.

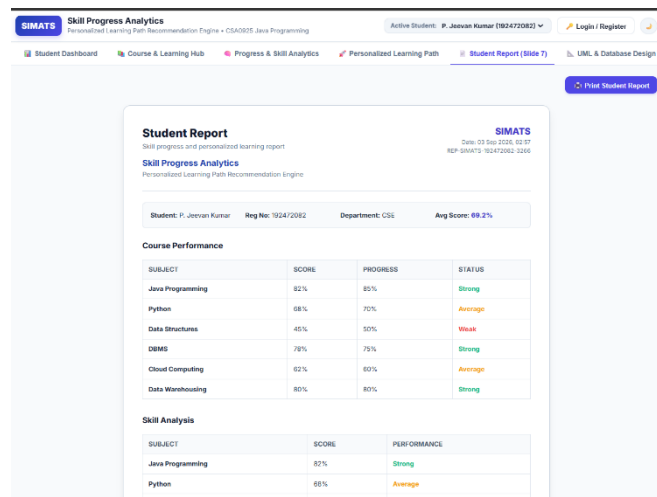


Fig 4:Student Report

REFERENCES

Use **IEEE referencing style** consistently throughout the report.

Your references should include sources for:

- Personalized learning
- Educational data mining
- Recommendation systems
- Student performance analytics
- Machine learning
- Learning analytics
- Software libraries used
- Dataset sources
- Applicable standards

For example, your final bibliography can contain categories such as:

1. Research papers on personalized learning.
2. Research papers on educational recommendation systems.
3. Research papers on student-performance prediction.
4. Documentation for Python/Pandas/scikit-learn.
5. Dataset source.
6. Applicable ISO/IEC standards.
7. Web resources used during implementation.
8. AI-assisted resources, if required by your institution's policy.

APPENDICES

Appendix A – Detailed Calculations

Include:

- Skill-score calculation
- Normalization calculation
- Recommendation-score calculation
- Statistical calculations
- ML evaluation calculations

Appendix B – Engineering Drawings / Schematics

Include:

- System architecture
- Data-flow diagram
- Recommendation workflow
- Database schema
- Application workflow

Appendix C – Source Code / Code Repository Information

Include:

- Project folder structure
- Important code excerpts
- Repository information
- API structure

Avoid putting the entire source code into the report.

Appendix D – Dataset Information

Include:

- Dataset description
- Attributes
- Number of records
- Data types
- Sample records

Appendix E – Experimental / Raw Data

Include:

- Test dataset
- Testing results
- Performance measurements
- Recommendation evaluation data

Appendix F – Ethics / Institutional Approval

Include this only if applicable.

Appendix G – Project Meeting / Progress Records

Include:

- Meeting dates
- Discussion points
- Decisions
- Progress photographs/signatures if required

Appendix H – User/Industry Feedback

Include:

- Feedback forms
- Survey results
- User testing

Appendix I – Publications / Patents / Competitions / Product Outcomes

Include only if applicable.

Appendix J – Individual Contribution Evidence

Include:

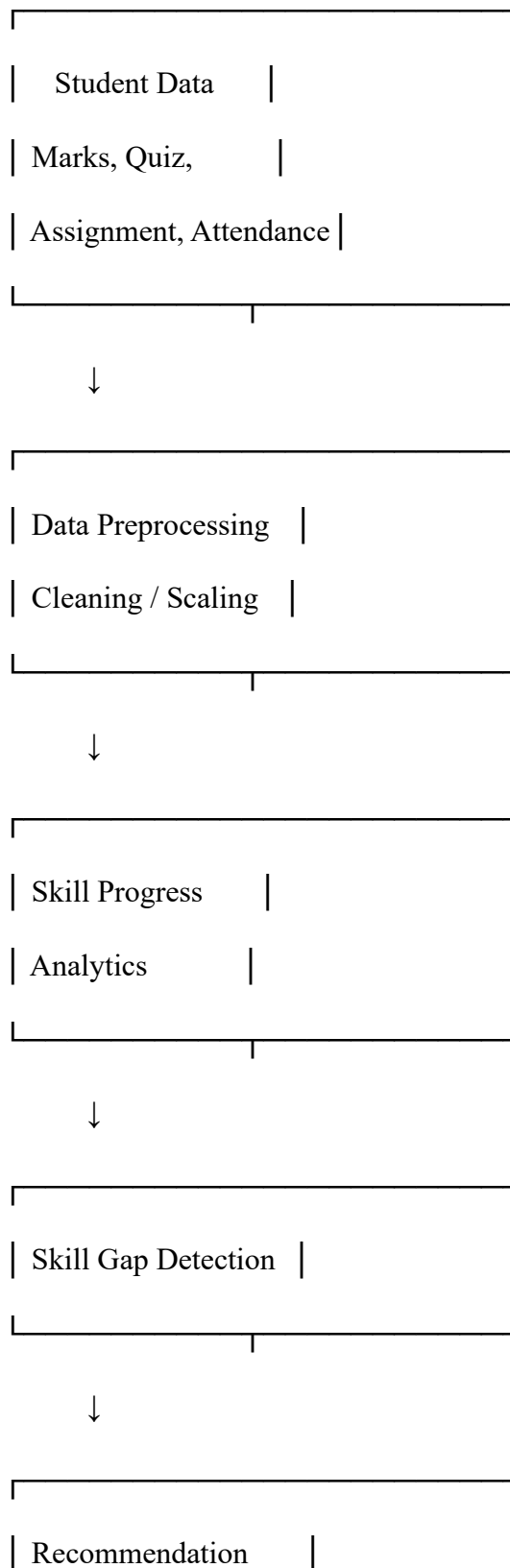
- Individual work
- GitHub commits
- Screenshots
- Module contribution
- Meeting records
- Testing contribution

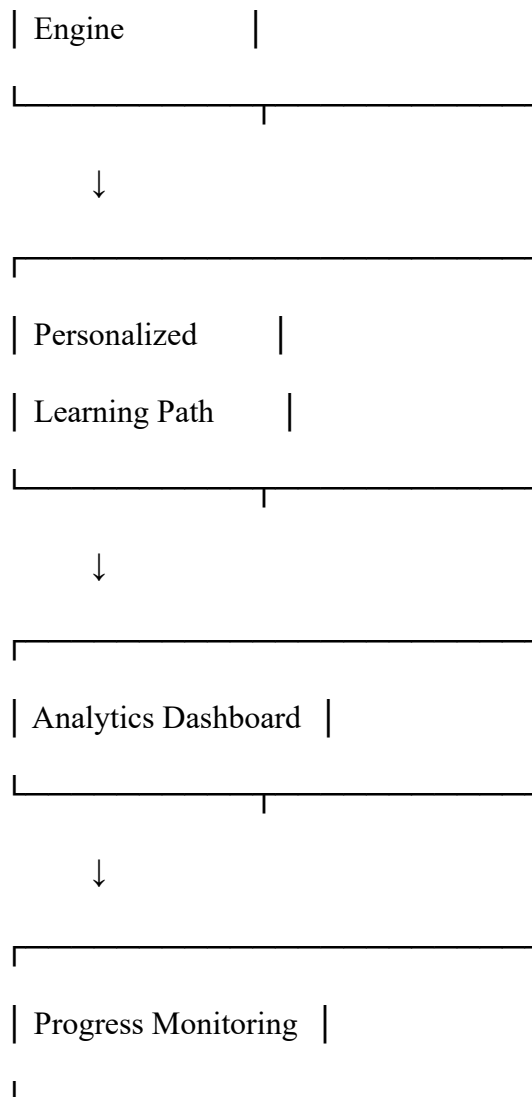
CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome / Area	Evidence	Location
Complex engineering problem formulation	Problem statement and requirements	Ch. 1
Engineering design under constraints	Design alternatives and trade-offs	Ch. 3
Technical communication	Complete report and presentation	Whole Report + Viva
Ethics and professional responsibility	Ethics, privacy and security	Ch. 3.7
Teamwork and leadership	Contribution and project planning	Ch. 4.8
Experimentation and engineering judgment	Testing and analysis	Ch. 4.3–4.6
Independent acquisition of knowledge	New technical skills	Ch. 5.3
Standards	Applicable standards	Ch. 3.2
Sustainability	Sustainability analysis	Ch. 3.7
Societal impact	Personalized education support	Ch. 3.7
Risk assessment	Risk register	Ch. 3.7
Security	Student-data protection	Ch. 3.7
Integrated/system-level engineering	System architecture	Ch. 3.5
Project management	Milestones, roles and budget	Ch. 4.8
Individual contribution	Individual evidence	Appendix J

Recommended final project architecture

For your particular title, I recommend using this overall architecture in the report:





Important: For Chapter 4, do not submit invented accuracy/results. Once you run your actual system, replace the “**Enter result**” fields with your measured values, screenshots, graphs, and test evidence. This will make the report much stronger during your capstone evaluation.